KV1200M040

1200 V, 40 mΩ silicon carbide power MOSFET, N-channel enhancement mode, TO-247-4 package

Features

- Low on-state resistance: 40 mΩ typical at V_{GS} = 18 V
- Low capacitance and gate charge for fast switching
- Kelvin source pin for clean gate drive
- Fast intrinsic body diode with low reverse recovery charge

Applications

Totem-pole power factor correction, solar inverters, EV chargers, industrial motor drives.

Absolute maximum ratings

T_C = 25 °C unless otherwise noted. Stresses beyond these ratings may damage the device.

Parameter	Symbol	Conditions	Value	Unit
Drain-source voltage	V_{DS}	V_{GS} = 0 V	1200	V
Gate-source voltage, maximum	$V_{GS,max}$	static	+22	V
Gate-source voltage, minimum	$V_{GS,min}$	static	−8	V
Continuous drain current	I_D	T_C = 25 °C	58	A
Continuous drain current	I_D	T_C = 100 °C	41	A
Pulsed drain current	$I_{D,pulse}$	pulse width limited by $T_{j,max}$	150	A
Power dissipation	P_D	T_C = 25 °C	288	W
Operating junction temperature	T_j		−55 to 175	°C
Storage temperature	T_{stg}		−55 to 150	°C

Thermal characteristics

Parameter	Symbol	Typ.	Max.	Unit
Thermal resistance, junction to case	$R_{\theta JC}$	0.38	0.52	K/W
Thermal resistance, junction to ambient	$R_{\theta JA}$		40	K/W

Electrical characteristics

T_j = 25 °C unless otherwise noted.

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Unit
Gate threshold voltage	$V_{GS(th)}$	V_{DS} = V_{GS}, I_D = 10 mA	1.8	2.6	3.6	V
Zero gate voltage drain current	I_{DSS}	V_{DS} = 1200 V, V_{GS} = 0 V		1	50	µA
Gate-source leakage current	I_{GSS}	V_{GS} = 22 V			100	nA
Drain-source on-state resistance	$R_{DS(on)}$	V_{GS} = 18 V, I_D = 30 A		40	52	mΩ
Drain-source on-state resistance	$R_{DS(on)}$	V_{GS} = 18 V, I_D = 30 A, T_j = 175 °C		68		mΩ
Input capacitance	C_{iss}	V_{DS} = 800 V, f = 1 MHz		1950		pF
Output capacitance	C_{oss}	V_{DS} = 800 V, f = 1 MHz		105		pF
Reverse transfer capacitance	C_{rss}	V_{DS} = 800 V, f = 1 MHz		9		pF
Total gate charge	Q_g	V_{DS} = 800 V, I_D = 30 A		106		nC

Switching characteristics

V_{DD} = 800 V, I_D = 30 A, V_{GS} = −4 V / +18 V, $R_{G(ext)}$ = 2.5 Ω, inductive load, T_j = 25 °C.

Parameter	Symbol	Typ.	Unit
Turn-on delay time	$t_{d(on)}$	14	ns
Rise time	t_r	22	ns

Turn-off delay time	$t_{d(off)}$	31	ns
Fall time	t_f	11	ns
Turn-on switching energy	E_{on}	410	µJ
Turn-off switching energy	E_{off}	95	µJ

Body diode characteristics

Parameter	Symbol	Conditions	Typ.	Unit
Diode forward voltage	V_{SD}	V_{GS} = −4 V, I_{SD} = 15 A	4.1	V
Reverse recovery time	t_{rr}	di/dt = 2000 A/µs	18	ns
Reverse recovery charge	Q_{rr}	di/dt = 2000 A/µs	160	nC

Notes

1. The recommended gate drive is −4 V for turn-off and +18 V for turn-on. A negative turn-off voltage below −4 V increases the threshold drift over the device lifetime.
2. $R_{DS(on)}$ rises with junction temperature by about 70% from 25 °C to 175 °C.
3. All values are typical of production devices and are not tested in production unless a minimum or maximum is given.

Ordering information

KV1200M040-T4: TO-247-4, tube of 30. KV1200M040-T3: TO-247-3, tube of 30, without the Kelvin source pin; its switching energies are higher.